Daniel W. Cranston

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Summary

Computer scientist and mathematician specializing in graph theory and algorithm design. Former university professor with 16 years of faculty experience, approximately 90 research papers, and a graduate textbook on graph coloring. Experienced in designing algorithms, proving mathematical results, and communicating technical ideas. Also experienced in implementing algorithms in Python and testing their correctness and performance on large collections of graphs.

Technical Strengths

Algorithm design: Graph algorithms, combinatorial optimization, structural graph theory, algorithm correctness and mathematical proofs.
Computational experiments: Python, NetworkX, algorithm implementation, correctness checks, runtime measurement.
Research and communication: Research publications, technical writing, graduate-level instruction, presentations, collaboration.

Selected Computational and Algorithmic Research

Research topics: Graph coloring, list coloring, correspondence coloring, recoloring and reconfiguration.
Methods: Kempe swaps, discharging, reducibility, the potential method.

Computational experiments: Implemented graph algorithms in Python using NetworkX; tested correctness, measured runtime, and evaluated performance across graph instances.

Professional Experience

Virginia Commonwealth University — Richmond, Virginia 2009–2025
Associate Professor of Computer Science (2020–2025); *Associate Professor of Mathematics* (2015–2020); *Assistant Professor of Mathematics* (2009–2015)

- Conducted independent research in graph theory, algorithms, combinatorics, and discrete optimization.
- Published approximately 90 research papers, including collaborative work with researchers at other institutions.
- Designed algorithms for graph-coloring problems and proved rigorous results about their correctness and applicability.
- Taught courses in algorithms, theory of computation, graph theory, combinatorics, multivariable calculus, linear algebra, introduction to proof, and related subjects.
- Mentored student projects that won awards within the department and the college of engineering.
- Communicated technical material to audiences with varied mathematical and computing backgrounds.

Johns Hopkins University 2016–2017
Visiting Associate Professor of Applied Mathematics and Statistics

DIMACS / Bell Labs 2007–2009
Postdoctoral Fellow — research in discrete mathematics and algorithms.

Education

University of Illinois Urbana-Champaign 1999–2007
Ph.D. in Computer Science, 2007; M.S. in Computer Science, 2004;
M.S. in Applied Mathematics (Optimization and Algorithms), 2001.

Greenville College 1996–1999
B.A. in Mathematics, magna cum laude, 1999.

Selected Publications and Technical Writing

- *Graph Coloring Methods* (2024). Graduate textbook, 453 pages. Freely available online.
- Nicolas Bousquet and Daniel W. Cranston, "A Linear Kernel for Independent Set Reconfiguration in Planar Graphs." *SIAM Journal on Discrete Mathematics*, to appear.
- Stijn Cambie, Wouter Cames van Batenburg, Daniel W. Cranston, Jan van den Heuvel, and Ross J. Kang, "Reconfiguration of List Colourings." Manuscript under review.
- Daniel W. Cranston, "List-Recoloring of Sparse Graphs." *European Journal of Combinatorics* 105 (2022), 103562.
- Daniel W. Cranston and Douglas B. West, "An Introduction to the Discharging Method via Graph Coloring." *Discrete Mathematics* 340 (2017), 766–793.

Full publication list: arXiv publication page.

Professional Service

Editor, *Electronic Journal of Combinatorics* (2025–present); Editorial Board, *Journal of Graph Theory* (2018–present) and *Discussiones Mathematicae Graph Theory* (2019–present). Refereed more than 130 research articles.